

MUHAMMAD RIZKY RAMADHANI

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PERSONAL INFORMATION

Full Name	Muhammad Rizky Ramadhani
Date of Birth	Tulungagung, 20 November 2002
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ORCID	0009-0004-7117-8954
Area of Expertise	Advanced Material Physics

PROFILE

Physics graduate (GPA 3.72/4.00) with a published Q1 journal article and a growing focus on data analytics — combining a rigorous scientific background with hands-on experience in quantitative data analysis, Python, and data visualization. Experienced in managing and interpreting multi-variable datasets from materials characterization, producing publication-ready figures, and communicating technical findings clearly. Actively developing business data skills through Excel, Python, and data analytics training. Two years of teaching experience across 8 technical laboratory courses and field leadership in a community development program.

EDUCATION

Bachelor of Science in Physics University of Jember	2021 – 2025
GPA 3.72/4.00	
Thesis: Effect of Polyvinyl Alcohol Concentration on Mechanical and Physical Properties of PVA/Chitosan Composite Films Reinforced with Coffee Husk Cellulose	
Supervisor: Prof. Dr. Edy Supriyanto, S.Si, M.Si.	

SKILLS & COMPETENCES

Data Analysis & Visualization: Python (NumPy, Pandas, Matplotlib, Seaborn) · Microsoft Excel (Pivot Table, Dashboard) · Origin by OriginLab

Programming Languages: Python · MATLAB · C++

Software & Tools: Microsoft Office (Word, Excel, Powerpoint, Outlook) · Google Workspace · Notion · Canva

Characterization — Operation: Universal Testing Machine (UTM) · SEM · XRD

Characterization — Data Interpretation: Mechanical Testing (tensile strength, elongation at break) · FTIR · XRD · SEM

Hard Skills: Scientific Writing & Literature Review · Research Methodology · English Proficiency (TOEFL ITP 563) · 10-Finger Typing

Soft Skills: Team Leadership · Project Management · Stakeholder Coordination · Technical Communication · Teaching & Mentoring · Problem Analysis

RESEARCH & PUBLICATIONS

- Research Author (Second Author) |** Materials Physics Laboratory, UNEJ January 2024 – February 2026
- Designed and fabricated PVA/Chitosan composite films reinforced with coffee husk cellulose across 40 experimental configurations using UTM, FTIR, XRD, and SEM.
 - Analyzed 6 characterization datasets (mechanical testing, FTIR, XRD, and SEM) using Origin and Python; produced 16 publication-ready figures and tables for the final manuscript.
 - Contributed to manuscript writing and revision across 3 submission rounds; research published in RSC Advances (Q1, 2026) — DOI: 10.1039/D5RA08990E.

Publication:
E. Supriyanto, **M. R. Ramadhani**, D. S. B. Prasetya, S. Akbar, S. Subekti, A. F. Sunartomo, dan K. Triyana, “Green Design of Biodegradable Packaging Films via Renewable Coffee Waste and PVA Matrix Tuning,” RSC Adv., 2026, 16, 16553–16563. <https://doi.org/10.1039/D5RA08990E>

TEACHING EXPERIENCE

- Laboratory Teaching Assistant |** Department of Physics, University of Jember August 2023 – August 2025
- Guided and evaluated approximately 30 students per semester across 8 technical laboratory courses over 4 consecutive semesters — Computational Physics I & II, Electronics I & II, Digital Engineering, Optoelectronics, Sensors & Transducers, and Electromagnetism.
 - Facilitated 16 laboratory sessions per semester; enforced standard operating procedures and safety compliance throughout.
 - Delivered technical explanations across diverse student cohorts, building fluency in communicating complex topics at different levels of familiarity

COMMUNITY SERVICE

- Team Leader, KKN PPM |** University of Jember · Kloposawit, Lumajang July 2024 – August 2024
- Led a team of 12 students in designing and executing 3 community development programs over 6 weeks.
 - Coordinated communication across 5 institutional stakeholders: LP2M UNEJ, local government officials, and community representatives.
 - Responsible for strategic decisions, resource allocation, and progress reporting to the university.

COMMITTEE EXPERIENCE

- Committee Member |** 2nd International Basic Science Conference (IBSC) 2024
- Committee Member |** Student Assembly (MAWA) — HIMAFI Physics Student Association 2021

CERTIFICATIONS & TRAINING

- Mini Bootcamp Introduction to Data Analytics (SQL, Python, Excel) Batch 12 |** DQLab 2026
- Data Science dengan Microsoft Fabric (METC) |** Microsoft Elevate Training Center × Dicoding 2026
- Mini Bootcamp Introduction to Software Engineering |** Revou 2026
- Program Terapan Bidang TIK (PRODISTIK) |** Sepuluh Nopember Institute of Technology 2021